

Victor Chua

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Singaporean Citizen | [linkedin.com/in/vicchua123](https://www.linkedin.com/in/vicchua123) | Internship: January – August 2027 | Open to remote
Applied AI Engineering | LLM Systems: RAG, Retrieval & On-Prem Inference | Evaluation Harnesses

Education

Nanyang Technological University (NTU) Aug 2024 – Dec 2028
Bachelor of Computing (Computer Science) with Turing AI Scholars Programme (TAISP)

Work Experience

SUSS Digital Solutions & AI Singapore
AI & Software Engineer Intern May 2026 – Aug 2026
Applied LLM Tooling, Evaluation & Internal Automation

- Built an **evaluation harness** for lecture diagnostics: faster-whisper transcription feeding **LLM-scored rubrics**, per-module eval JSONs, and a composite metric ranking delivery quality across the cohort
- Built an invoice pipeline on a locally hosted **Llama 3 3B** so financial data never left the network; a **human approval gate** and digital signature precede send, cutting a 10-minute task to one click over 50+ invoices
- Automated lecture quality review with **FastAPI** over the **TwelveLabs** multimodal API, which judges engagement semantically rather than just silence, returning trimmed footage and flagged timestamps from 3-hour lectures
- Co-presented AI tool evaluations and cost-benefit analyses to **non-technical senior leadership**, translating deployment decisions into business terms and addressing reliability and accuracy concerns

Yap Motor Singapore
Machine Learning Intern Dec 2025 – Jan 2026
First Production AI System — Retrieval-Augmented Generation

- Built Yap Motor's first AI system — a production **RAG** pipeline (**Google Gemini** + **ChromaDB**) letting 24 mechanics query 500+ pages of multilingual repair manuals; still in use 6 months after the internship
- Reached **90% retrieval precision** across 8 diagnostic categories by iterating on chunk size, overlap, and embedding strategy, validated against a month of structured mechanic feedback on live repair queries
- Engineered a multi-stage retrieval pipeline: folder-based priority scoring, **regex NLP** pulling vehicle make/model/year from natural language, dynamic ChromaDB filters, and **re-ranking** before LLM reasoning
- Built a multilingual (EN/CN/JP) ingestion pipeline for PDFs and Word docs of variable structure; debugged two production blockers — ChromaDB schema validation failures and case-sensitive metadata filters

DBS Singapore
Data Engineer Intern May 2023 – May 2024
Enterprise Pipeline Monitoring & ML Feature Retrieval

- Built a **Grafana/SQL** dashboard tracking RAM and execution metrics across every version of a 3,000+ job/day pipeline; surfaced version-specific bottlenecks that teams then fixed for 20% RAM and 15% speed gains
- Developed a **REST client** integrating **Elasticsearch** into the team's internal Python library, giving production ML models real-time feature retrieval through shared infrastructure for the first time

Personal Projects

Drum Hub – AI drum-score reader | hf.co/Icyautumn/drum-omr-mobilenetv3 Jun 2026 – Present

- Trained a dual-head **MobileNetV3** optical music recognition model (2.47M params) by two-phase transfer learning: 90.9% duration accuracy, 0.93/0.89 kick/snare F1 on 1,924 held-out bars from 26 unseen songs
- Exported the trained **PyTorch** model to **ONNX** and published the weights and ONNX artifact under MIT, so the score reader runs offline on a student's machine with no network call or per-use API cost
- Curated the first publicly available labelled drum notation dataset — 19,948 bar images from 264 songs across 19 drum types and 10 duration classes, split by song to prevent leakage and QC'd over 20+ sampling rounds
- Built the score editor in **TypeScript** with **VexFlow 5** inside an **Electron** desktop app: correct voice-split stem direction logic and note-level editing from rest through 32nd notes with beam linkages

Leadership & Activities

TAISP x MLDA Workshop Lead, NTU School of Computing 2025
Designed and delivered a hands-on PyTorch workshop curriculum with teaching notebooks for NTU undergraduates, taking students from tensor basics to a trained image classifier in a single session.

Technical Skills

Languages: Python, TypeScript, SQL, JavaScript

AI & LLM Systems: RAG, ChromaDB, Vector Search, Chunking Strategy, Re-ranking, On-Premises Inference (Llama 3 3B), LLM APIs (Gemini, GPT, Claude), Prompt Design, Structured Output Extraction

Evaluation & Reliability: LLM-as-Judge Harnesses, Rubric & Composite Scoring, Retrieval Precision, F1 / Accuracy Benchmarking, Human-in-the-Loop Approval Gates

ML & Deployment: PyTorch, Transfer Learning, MobileNetV3, ONNX Runtime, HuggingFace, faster-whisper, TwelveLabs

Full-Stack & Data: Electron, VexFlow 5, FastAPI, REST API Design, Elasticsearch, Grafana, Git, Document Ingestion